Aileron Angle Position Indicator – General data

1. General

A potentiometer is mounted on the port side aileron hinge arm at station W 247.35 and sends a signal of the aileron angle position through the DAU and onto the Flight Data Recorder (if installed).

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent.

A. Remove the Aileron Angle Position Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Disconnect the electrical connection at the terminal block.

4) Loosen the nut and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket. Remove the lever assembly.

5) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

6) Remove potentiometer.

B. Install the Aileron Angle Position Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Position the mounting bracket on aileron hinge arm and secure in place with (3) bolts, washers and nuts.

5) Locate the lever assembly on potentiometer shaft leaving a gap of 0.02 to 0.25 inch between the lever assembly and the potentiometer housing to minimize the push rod assembly misalignment.

6) Secure with a washer, nut and screw.

7) Install bolt through the push rod assembly end, install washer, and then bolt through the lever assembly end. Install washer, nut and cotter pin.

8) Adjust the aileron angle position potentiometer. (Refer to Para 3.A).

9) Reconnect the electrical connection at the terminal block.

10) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

3. Adjustment/Test

A. Adjustment Aileron Angle Position Potentiometer

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clips.

4) Move flaps to the full up position.

5) Move ailerons to the neutral position and lock in place with aileron rigging pin.

6) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

7) Loosen the screw and nut on the lever assembly at the potentiometer shaft enough that the shaft turns freely.

8) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker and monitor DC voltage from terminal 2 to aircraft ground.

9) Rotate potentiometer shaft until DC Voltage is 50% ± 2% of Ground Power Voltage (V_{GP}) (approximately 14 volts).

10) Reconnect pushrod and lever assembly to potentiometer shaft.

Note Do not rotate potentiometer shaft during this process.

11) Tighten screw and nut on the lever assembly leaving a gap of 0.02 to 0.05 of an inch between lever assembly and potentiometer housing.

12) Confirm voltmeter reading is unchanged.

13) Remove aileron rigging pin.

14) With voltmeter connected, function test aileron control position per the following:

a) Deflect ailerons to the full up position (17.5 deg) and confirm DC Voltage is 20.8% ± 2.2% of Ground Power Voltage (V_{GP}).

b) Deflect ailerons to the full down position (16 deg) and confirm DC Voltage is 76.7% ± 2.2% of Ground Power Voltage (V_{GP}).

15) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

16) Remove voltmeter and reconnect terminal 2 of the potentiometer electrical connections at the terminal block.

17) Remove safety clips and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

18) Deflect the ailerons full left and ensure DAU Maintenance Status CAS message does not activate.

19) Deflect the aileron full right and ensure DAU Maintenance Status CAS message does not activate.

20) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Aileron Position Voltage Check

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clips.

4) Move flaps to the full up position.

5) Move ailerons to the neutral position and lock in place with aileron rigging pin.

6) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.

7) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker and monitor DC voltage from terminal 2 to aircraft ground

8) If DC Voltage is not 50% ± 2% of V_{GP}, do adjustment steps per Para 3.A.

9) Remove aileron rigging pin.

10) With voltmeter connected, function test aileron control position per the following:

a) Deflect ailerons to the full up position (17.5 deg) and confirm DC Voltage is 20.8% ± 2.2% of Ground Power Voltage (V_{GP}).

b) Deflect ailerons to the full down position (16 deg) and confirm DC Voltage is 76.7% ± 2.2% of Ground Power Voltage (V_{GP}).

11) If DC Voltage does not correspond, do adjustment steps per Para 3.A.

12) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

13) Remove voltmeter and reconnect terminal 2 of the potentiometer electrical connections at terminal block.

14) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

15) Deflect the ailerons full left and ensure DAU Maintenance Status CAS message does not activate.

16) Deflect the ailerons full right and ensure DAU Maintenance Status CAS message does not activate.

17) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.